



Fig. 4: Threat landscape of attacks on smartphone sensors

(synthetically crafted or otherwise) into a system in a way that is indistinguishable with regards to real data, thereby, poisoning the rest of the sensor data ingestion pipeline. A classic example of this would be a GPS spoof attack where a software-defined radio device is engineered to synthesise fake GPS signals that are ingested by the targeted device as if these were the GPS signals beamed from satellites.

Recently, a US\$ 1 million grant was awarded to Clemson University researchers to counter GPS spoofing attacks on network time protocol (NTP). Rooted or jail-broken devices are also targeted to carry out spoof injection attacks where synthetic gyroscopic, accelerometer and magnetometric (compass) data is injected as if these emanated from the physical sensors (think cheating in fitness apps or mobile games).

Spoofing by transduction attacks involve targeting the very physical environment being sensed by the sensor with a malicious intent. The two motion sensors—accelerometer and gyroscope—can both be targeted acoustically by strategically playing certain sound frequencies that induce maximum sensing response (resonance) from the respective MEMS chips. These sound signals can either emanate from a pre-calibrated sonic gun, a nearby laptop playing YouTube videos or a silent partner-in-crime that resides on the device, the vibrator motor.

With regards to the vibrator motor specifically, one very interesting development that will pave the way for exciting enhancements in the richness of haptic interactions, which might also contribute to increased sophistication of security threats, is the latest Android API (level 26) that allows for triggering of the vibrator motor via much complex waveforms with precise control over duration, waveform shape as well as duty cycle.

Malicious use of sensor data. This category of attacks entails the attacker munging sensor data to infer aspects of the person's or even the device's identity with a malicious intent.

As was revealed in a recent study, a significant number of smartphone users had never even heard of the accelerometer MEMS chip present on their device, and were relatively much less concerned about the malicious use of these sensors in comparison with overtly active biometric sensors such as fingerprint or camera

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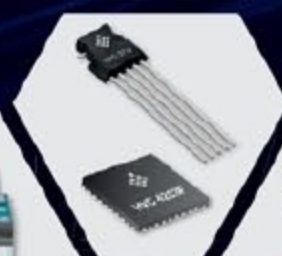


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